

Candidate 1 — LTM-Port Saturation, Communication Concentration and Delay

Complete Structured Experimental Notes

1. qcomm — What Is It?

- qcomm is a simulator/framework for studying communication in modular multi-core quantum systems.
- The system contains multiple QPUs/cores connected through an interconnect.
- When a two-qubit gate involves qubits located on different QPUs, inter-QPU communication is required.
- Each QPU has a limited number of LTM (Long-Term Memory) ports that provide communication parallelism.

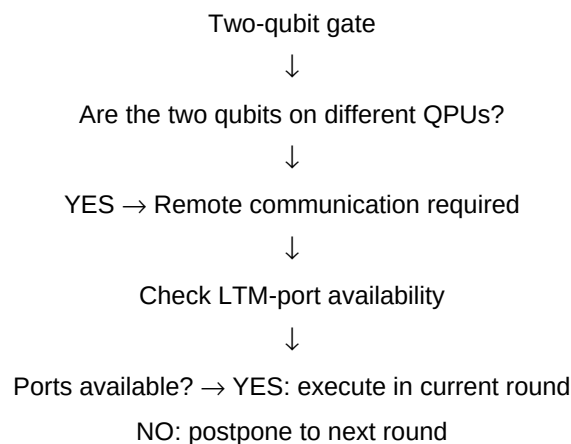
Basic system view

Multiple QPUs → Interconnect → Remote quantum communication → LTM-port resource

2. How qcomm Handles Remote Communication

When a remote two-qubit operation is encountered, qcomm determines the source and destination QPUs and checks the available LTM ports.

Flow of a remote communication



Thus, LTM ports determine how many communication operations can be handled in parallel during a scheduling round.

3. Important qcomm Constraints

3.1 LTM-port constraint

A QPU has a finite number of LTM ports. If simultaneous communication demand exceeds the available ports, not all communications can be handled in the same round.

3.2 Communication-round constraint

qcomm schedules remote communications in parallel rounds. Communications that cannot fit because of port availability are postponed to later rounds.

3.3 QPU qubit-capacity constraint

Each QPU has a maximum logical-qubit capacity. Very large or poorly balanced migrations can therefore cause a QPU-capacity/mapping failure.

3.4 Mapping constraint

After an accepted remote operation, qcomm updates the qubit-to-QPU mapping. Therefore, simply adding a two-qubit gate does not guarantee that every added gate remains a remote communication.

4. Why Did We Investigate Saturation After 3 Ports?

The qcomm study reports that no further communication-time improvement was observed beyond 3 LTM ports for the workloads tested in that study.

Research question: Is 3 LTM ports an inherent limit of qcomm, or is the observed saturation caused by the communication pattern of the tested workloads?

Two possible explanations

3 ports are an architectural limit



Ports above 3 cannot provide additional benefit

OR

The tested workloads did not create more than 3 simultaneous demands on the same QPU



Therefore, additional ports were not needed

5. Experiment 1 — LTM-Port Saturation

We constructed controlled workloads A, B, C, D and F and swept the number of LTM ports. The main purpose was to vary the concentration of remote communications across QPUs.

Important distinction

Total communication load = total number of remote communications.

Communication concentration = how many communications simultaneously involve the same QPU.

Because LTM ports are a per-QPU resource, concentration is important for determining whether a QPU runs out of ports.

5.1 Workload A

1 remote communication; 8 qubits and 9 gates.

LTM Ports	Remote Rounds	Teleportation Time (μ s)	Execution Time (μ s)
1	1	1.43001	1.75576
2	1	1.43001	1.75576
3	1	1.43001	1.75576
4	1	1.43001	1.75576

Observation: One port is already sufficient. Increasing the port count provides no additional benefit.

5.2 Workload B

2 remote communications; 8 qubits and 10 gates.

LTM Ports	Remote Rounds	Teleportation Time (μ s)	Execution Time (μ s)
1	2	2.86002	3.25089
2	1	1.43001	1.78089
3	1	1.43001	1.78089
4	1	1.43001	1.78089

Observation: With one port, the two communications require two rounds. Two ports allow one-round execution; additional ports provide no further benefit.

5.3 Workload C

3 remote communications; 12 qubits and 15 gates.

LTM Ports	Remote Rounds	Teleportation Time (μ s)	Execution Time (μ s)
1	3	4.30003	4.89634
2	2	2.87002	3.42633
3	1	1.44001	1.95632
4	1	1.44001	1.95632

Observation: Three ports are required to execute all three communications in one round.

5.4 Workload D — Critical 4-Port Result

4 remote communications concentrated at one destination QPU; 16 qubits and 20 gates.

LTM Ports	Remote Rounds	Teleportation Time (μ s)	Execution Time (μ s)
1	4	5.73004	6.51179
2	2	2.87002	3.57177
3	2	2.87002	3.57177
4	1	1.45001	2.11176
5	1	1.45001	2.11176
6	1	1.45001	2.11176

Observation: At 3 ports, the workload still requires two communication rounds. At 4 ports, it fits into one round. Ports 5 and 6 provide no further improvement.

CRITICAL RESULT: Workload D saturates at 4 LTM ports.

5.5 Workload F — Distributed 4-Communication Case

4 remote communications distributed across QPUs; 16 qubits and 20 gates.

LTM Ports	Remote Rounds	Teleportation Time (μ s)	Execution Time (μ s)
1	2	2.87002	3.59177
2	1	1.46001	2.14176
3	1	1.46001	2.14176
4	1	1.46001	2.14176
5	1	1.46001	2.14176

Observation: Although F also has four remote communications, two ports are sufficient because the communications are distributed across QPUs.

6. Cross-Workload Saturation Results

Workload	Remote Communications	Saturation Point
A	1	1 port
B	2	2 ports
C	3	3 ports
D	4	4 ports
F	4	2 ports

Final result of Experiment 1: We obtained a valid workload that requires 4 LTM ports for one-round execution.

Therefore, the 3-port saturation observed in the original qcomm experiments is **not a universal property of qcomm**.

7. D vs F — Same Load, Different Saturation

This is the strongest comparison in the saturation experiment.

Property	Workload D	Workload F
Remote communications	4	4
Communication pattern	Concentrated	Distributed
Saturation point	4 ports	2 ports

Workload D — concentrated

QPU 0 ← communication

QPU 0 ← communication

QPU 0 ← communication

QPU 0 ← communication

Peak demand = 4

Workload F — distributed

QPU 0 ← communication

QPU 1 ← communication

QPU 2 ← communication

QPU 3 ← communication

Demand is distributed across QPUs

Main observation: D and F have the same total communication load, but their LTM-port requirements are different.

Therefore, communication concentration matters, not total communication load alone.

8. Experiment 2 — Communication Delay with Fixed 3 Ports

After establishing that LTM-port requirements depend on communication concentration, we fixed the LTM-port count at 3 and increased the number of remote communications.

Question: What happens when communication demand exceeds the available 3-port parallelism?

Remote Communications	Rounds	Execution Time (μ s)
1	1	2.35109
2	1	2.37873
3	1	2.41637
4	2	3.91402
5	2	3.94166
6	2	3.97930
7	3	5.47695
8	3	5.50459
9	3	5.54223

9. Communication-Round Pattern

1–3 communications → 1 round

4–6 communications → 2 rounds

7–9 communications → 3 rounds

Interpretation: Once the communication demand exceeds the available parallelism, the excess communication is moved to additional scheduling rounds.

10. Execution-Delay Observation

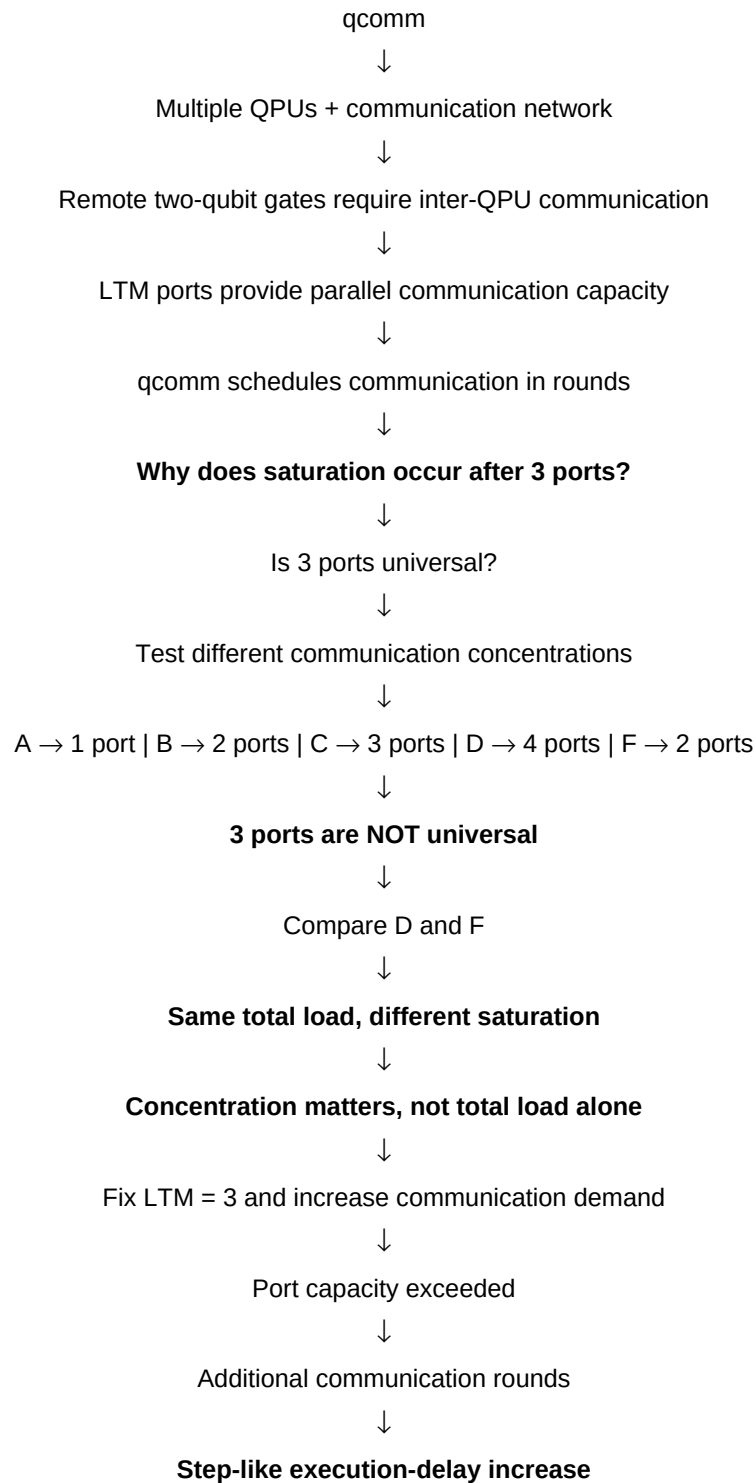
Transition	Execution-time change
3 → 4 communications	+1.49765 μ s
6 → 7 communications	+1.49765 μ s
1 → 2 communications	+0.02764 μ s
2 → 3 communications	+0.03764 μ s
4 → 5 communications	+0.02764 μ s
5 → 6 communications	+0.03764 μ s
7 → 8 communications	+0.02764 μ s
8 → 9 communications	+0.03764 μ s

Key pattern: Adding communication while remaining in the same number of rounds causes only a small increase in execution time. When the additional communication forces a new round, a much larger execution-time jump occurs.

3 → 4: 1 round → 2 rounds → +1.49765 μ s

6 → 7: 2 rounds → 3 rounds → +1.49765 μ s

11. Complete Research Flow



12. Final Observations

- The 3-LTM-port saturation reported in qcomm is **not universal**.
- Our controlled workload D requires **4 LTM ports** for one-round execution.
- Therefore, a fourth LTM port can provide additional benefit for a suitable workload.
- Workloads D and F both contain 4 remote communications, but D saturates at 4 ports while F saturates at 2 ports.
- Therefore, **communication concentration matters more than total communication load alone** for LTM-port adequacy in these experiments.

- When communication demand exceeds available LTM-port parallelism, qcomm postpones excess communication to later rounds.
- Increasing communication within an existing round causes a small delay increase.
- Forcing an additional communication round causes a much larger delay increase.
- In the tested fixed-3-port configuration, an additional forced round added approximately **1.49765 μ s** to execution time.

13. Final Candidate-1 Statement

The observed 3-LTM-port saturation in qcomm is workload-dependent rather than a universal architectural limit. Our controlled experiments show that LTM-port requirements depend on the concentration of simultaneous communication demand across QPUs: a concentrated workload required 4 ports, while another workload with the same total communication load required only 2. When communication demand exceeds the available 3-port parallelism, qcomm serializes the excess communication into additional rounds, producing a step-like increase in execution delay.

Scientific status: These are controlled empirical observations. They do not yet establish a universal mathematical formula for minimum LTM-port count or a universal delay penalty for every circuit, mapping, or architecture.